# XLPE CABLE - SINGLE CORE

Single Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

## Physical Data

| Nominal Area | Minium No. of Strand in Conductor | | Insulation Thickness | | Armour Wire Dimension | | Outer Sheath Thickness | | Approx. Overall Diameter | | Approx. Net Weight of the Cables | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Arm | Un-Arm | Round | Formed | Arm | Un-Arm | Arm | Un-Arm | Armoured | | UnArmoured | |
| Sq. mm | Aluminium Nos | Copper Nos | mm | mm | mm | mm | mm | mm | mm | mm | Kg/Km Al | Kg/Km Cu | Kg/Km Al | Kg/Km Cu |
| 4 | 1/3 | 1/7 | 1.00 | 0.70 | 1.40 | -- | 1.24 | 1.80 | 10.00 | 8.00 | 130 | 155 | 75 | 100 |
| 6 | 1/3 | 1/7 | 1.00 | 0.70 | 1.40 | -- | 1.24 | 1.80 | 11.00 | 9.00 | 145 | 180 | 90 | 120 |
| 10 | 1/7 | 1/7/6 | 1.00 | 0.70 | 1.40 | -- | 1.24 | 1.80 | 11.50 | 9.50 | 170 | 230 | 105 | 165 |
| 16 | 6 | 6 | 1.00 | 0.70 | 1.40 | -- | 1.24 | 1.80 | 12.50 | 10.50 | 200 | 295 | 130 | 225 |
| 25 | 6 | 6 | 1.20 | 0.90 | 1.40 | -- | 1.24 | 1.80 | 14.00 | 12.00 | 260 | 410 | 180 | 325 |
| 35 | 6 | 6 | 1.20 | 0.90 | 1.40 | -- | 1.24 | 1.80 | 15.00 | 13.00 | 305 | 515 | 215 | 425 |
| 50 | 6 | 6 | 1.30 | 1.00 | 1.40 | -- | 1.24 | 1.80 | 16.50 | 14.50 | 365 | 650 | 265 | 550 |
| 70 | 12 | 12 | 1.40 | 1.10 | 1.40 | -- | 1.24 | 1.80 | 18.50 | 16.50 | 460 | 870 | 350 | 760 |
| 95 | 15 | 15 | 1.40 | 1.10 | -- | 4.0x0.8 | 1.40 | 1.80 | 19.50 | 18.00 | 515 | 1080 | 440 | 1005 |
| 120 | 15 | 18 | 1.50 | 1.20 | -- | 4.0x0.8 | 1.40 | 1.80 | 21.50 | 19.50 | 610 | 1330 | 530 | 1245 |
| 150 | 15 | 18 | 1.70 | 1.40 | -- | 4.0x0.8 | 1.40 | 2.00 | 23.00 | 21.50 | 715 | 1605 | 650 | 1535 |
| 185 | 30 | 30 | 1.90 | 1.60 | -- | 4.0x0.8 | 1.40 | 2.00 | 25.00 | 24.00 | 865 | 1965 | 790 | 1890 |
| 240 | 30 | 34 | 2.00 | 1.70 | -- | 4.0x0.8 | 1.40 | 2.00 | 28.00 | 26.50 | 1070 | 2505 | 985 | 2415 |
| 300 | 30 | 34 | 2.10 | 1.80 | -- | 4.0x0.8 | 1.56 | 2.00 | 30.00 | 28.50 | 1300 | 3115 | 1180 | 2995 |
| 400 | 53 | 53 | 2.40 | 2.00 | -- | 4.0x0.8 | 1.56 | 2.20 | 34.00 | 32.50 | 1610 | 3945 | 1495 | 3835 |
| 500 | 53 | 53 | 2.60 | 2.20 | -- | 4.0x0.8 | 1.56 | 2.20 | 37.50 | 36.00 | 1980 | 4970 | 1860 | 4845 |
| 630 | 53 | 53 | 2.80 | 2.40 | -- | 4.0x0.8 | 1.72 | 2.20 | 41.50 | 40.00 | 2475 | 6360 | 2930 | 6195 |
| 800 | 53 | 53 | 3.10 | 2.60 | -- | 4.0x0.8 | 1.72 | 2.40 | 47.00 | 45.50 | 3095 | 8055 | 2930 | 7890 |
| 1000 | 53 | 53 | 3.30 | 2.80 | -- | 4.0x0.8 | 1.88 | 2.60 | 52.00 | 50.50 | 3820 | 9885 | 3645 | 9710 |

## Electrical Data

| Nominal Conductor Data | Max. DC Resisance of Conductor at 20°C | | Approx. AC Resistance of Max Operating Temp 90°C | | Appox. Capacitance | | Aprox. Reactance at 50 Hz | | Current Rating | | | | | | Short Circuit Rating for 1 Sec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | In Ground | | In Duct | | In Air | | | |
| Sq. mm | Ω/Km Al | Ω/Km Cu | Ω/Km Al | Ω/Km Cu | Un-Arm µF/Km | Arm µF/Km | Un-Arm Ω/Km | Arm Ω/Km | Amps Al | Amps Cu | Amps Al | Amps Cu | Amps Al | Amps Cu | kA (rms) Al | kA (rms) Cu |
| 4 | 7.4100 | 4.6100 | 9.4848 | 5.9008 | 0.29 | | 0.132 | | 37 | 47 | 34 | 43 | 33 | 41 | 0.38 | 0.57 |
| 6 | 4.6100 | 3.0800 | 5.9008 | 3.9424 | 0.34 | | 0.123 | | 47 | 58 | 43 | 53 | 43 | 52 | 0.56 | 0.86 |
| 10 | 3.0800 | 1.8300 | 3.9424 | 2.3424 | 0.43 | 0.32 | 0.114 | 0.134 | 59 | 77 | 54 | 70 | 55 | 71 | 0.94 | 1.43 |
| 16 | 1.9100 | 1.1500 | 2.4448 | 1.472 | 0.51 | 0.38 | 0.108 | 0.125 | 76 | 98 | 69 | 89 | 72 | 94 | 1.50 | 2.29 |
| 25 | 1.2000 | 0.7270 | 1.5360 | 0.93056 | 0.49 | 0.38 | 0.103 | 0.120 | 98 | 126 | 89 | 114 | 98 | 126 | 2.35 | 3.58 |
| 35 | 0.8680 | 0.5240 | 1.1110 | 0.67072 | 0.57 | 0.44 | 0.099 | 0.140 | 116 | 150 | 106 | 136 | 119 | 154 | 3.29 | 5.01 |
| 50 | 0.6410 | 0.3870 | 0.8205 | 0.49536 | 0.58 | 0.46 | 0.094 | 0.108 | 137 | 177 | 124 | 160 | 145 | 187 | 4.70 | 7.15 |
| 70 | 0.4400 | 0.2680 | 0.5632 | 0.34304 | 0.63 | 0.51 | 0.090 | 0.102 | 168 | 216 | 151 | 195 | 185 | 238 | 6.58 | 10.01 |
| 95 | 0.3200 | 0.1930 | 0.4096 | 0.24704 | 0.73 | 0.59 | 0.087 | 0.100 | 202 | 260 | 181 | 233 | 235 | 303 | 8.93 | 13.59 |
| 120 | 0.2530 | 0.1530 | 0.3238 | 0.19584 | 0.74 | 0.61 | 0.084 | 0.097 | 230 | 295 | 206 | 264 | 276 | 354 | 11.28 | 17.16 |
| 150 | 0.2060 | 0.1240 | 0.2637 | 0.15872 | 0.73 | 0.61 | 0.084 | 0.094 | 256 | 329 | 229 | 294 | 314 | 403 | 14.10 | 21.45 |
| 185 | 0.1640 | 0.0991 | 0.2099 | 0.126848 | 0.69 | 0.59 | 0.084 | 0.093 | 290 | 371 | 258 | 330 | 366 | 468 | 17.39 | 26.46 |
| 240 | 0.1250 | 0.0754 | 0.1600 | 0.096512 | 0.74 | 0.64 | 0.081 | 0.090 | 335 | 427 | 298 | 379 | 434 | 553 | 22.56 | 34.32 |
| 300 | 0.1000 | 0.0601 | 0.1280 | 0.076928 | 0.80 | 0.69 | 0.080 | 0.088 | 376 | 477 | 333 | 422 | 500 | 634 | 28.20 | 42.90 |
| 400 | 0.0778 | 0.0470 | 0.0996 | 0.06016 | 0.83 | 0.70 | 0.079 | 0.087 | 429 | 537 | 378 | 473 | 589 | 737 | 37.60 | 57.20 |
| 500 | 0.0605 | 0.0366 | 0.0774 | 0.046848 | 0.83 | 0.71 | 0.078 | 0.086 | 485 | 598 | 426 | 525 | 658 | 844 | 47.00 | 71.50 |
| 630 | 0.0469 | 0.0283 | 0.0600 | 0.036224 | 0.87 | 0.75 | 0.079 | 0.084 | 546 | 661 | 477 | 578 | 793 | 961 | 59.22 | 90.09 |
| 800 | 0.0367 | 0.0221 | 0.0470 | 0.028288 | 0.92 | 0.78 | 0.076 | 0.083 | 608 | 721 | 528 | 626 | 907 | 1077 | 75.20 | 114.40 |



# XLPE CABLE - TWO CORE

Two Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

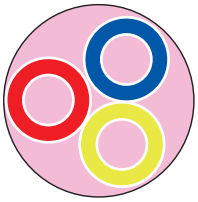
## Physical Data

| Nominal Area | Minium No. of Strand in Conductor | | Insulation Thickness (Nominal) | Minimum inner sheath thickness | Armour Wire Dimension | | Outer Sheath Thickness | | Approx. Overall Diameter | | Approx. Net Weight of the Cables | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Round | Formed | Arm | Un-Arm | Arm | Un-Arm | Armoured | | UnArmoured | |
| Sq. mm | Aluminium | Copper | mm | mm | mm | mm | mm | mm | mm | mm | Kg/Km | Kg/Km | Kg/Km | Kg/Km |
| | Nos | Nos | | | | | | | | | Al | Cu | Al | Cu |
| 4 | 1/3 | 1/7 | 0.70 | 0.30 | 1.40 | -- | 1.24 | 1.80 | 14.50 | 13.00 | 405 | 450 | 145 | 195 |
| 6 | 1/3 | 1/7 | 0.70 | 0.30 | 1.40 | -- | 1.24 | 1.80 | 16.00 | 14.00 | 455 | 525 | 175 | 245 |
| 10 | 1/7 | 1/7/6 | 0.70 | 0.30 | 1.40 | -- | 1.24 | 1.80 | 17.50 | 16.00 | 540 | 660 | 215 | 335 |
| 16 | 6 | 6 | 0.70 | 0.30 | 1.40 | -- | 1.40 | 1.80 | 16.50 | 15.00 | 555 | 750 | 250 | 440 |
| 25 | 6 | 6 | 0.90 | 0.30 | -- | 4.0X0.8 | 1.40 | 2.00 | 18.50 | 18.00 | 575 | 875 | 355 | 655 |
| 35 | 6 | 6 | 0.90 | 0.30 | -- | 4.0x0.8 | 1.40 | 2.00 | 20.00 | 19.50 | 685 | 1110 | 440 | 865 |
| 50 | 6 | 6 | 1.00 | 0.30 | -- | 4.0x0.8 | 1.40 | 2.00 | 22.00 | 21.50 | 810 | 1380 | 545 | 1115 |
| 70 | 12 | 12 | 1.10 | 0.30 | -- | 4.0x0.8 | 1.56 | 2.00 | 25.50 | 24.50 | 1040 | 1860 | 710 | 1535 |
| 95 | 15 | 15 | 1.10 | 0.40 | -- | 4.0x0.8 | 1.56 | 2.20 | 28.00 | 27.50 | 1275 | 2425 | 930 | 2070 |
| 120 | 15 | 18 | 1.20 | 0.40 | -- | 4.0x0.8 | 1.56 | 2.20 | 30.50 | 30.00 | 1500 | 3945 | 1110 | 2555 |
| 150 | 15 | 18 | 1.40 | 0.40 | -- | 4.0x0.8 | 1.72 | 2.20 | 33.00 | 32.00 | 1750 | 3540 | 1320 | 3105 |
| 185 | 30 | 30 | 1.60 | 0.50 | -- | 4.0x0.8 | 1.72 | 2.40 | 36.00 | 35.50 | 2085 | 4310 | 16445 | 3970 |
| 240 | 30 | 34 | 1.70 | 0.50 | -- | 4.0x0.8 | 1.88 | 2.60 | 40.50 | 40.00 | 2585 | 5475 | 2085 | 4975 |
| 300 | 30 | 34 | 1.80 | 0.60 | -- | 4.0x0.8 | 2.04 | 2.80 | 43.50 | 43.50 | 3100 | 6770 | 2555 | 6220 |
| 400 | 53 | 53 | 2.00 | 0.60 | -- | 4.0x0.8 | 2.36 | 3.00 | 49.50 | 59.00 | 3860 | 8580 | 3195 | 9715 |
| 500 | 53 | 53 | 2.20 | 0.70 | -- | 4.0x0.8 | 2.52 | 3.40 | 54.00 | 54.00 | 4725 | 10755 | 4055 | 10085 |
| 630 | 53 | 53 | 2.40 | 0.70 | -- | 4.0x0.8 | 2.68 | 3.60 | 60.00 | 60.00 | 5775 | 13635 | 5035 | 12895 |

## Electrical Data

| Nominal Conductor Data | Max. DC Resisance of Conductor at 20°C | | Approx. AC Resistance of Max Operating Temp 90°C | | Appox. Capacitance | Aprox. Reactance at 50 Hz | Current Rating | | | | | | Short Circuit Rating for 1 Sec | | Volrage Drop | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | In Ground | | In Duct | | In Air | | | | | |
| Sq. mm | Ω/Km | Ω/Km | Ω/Km | Ω/Km | | | Amps | Amps | Amps | Amps | Amps | Amps | kA (rms) | | V/A/Km | |
| | Al | Cu | Al | Cu | µF/Km | Ω/Km | Al | Cu | Al | Cu | Al | Cu | Al | Cu | Al | Cu |
| 4 | 7.4100 | 4.6100 | 9.4848 | 5.9008 | 0.220 | 0.093 | 42 | 54 | 36 | 45 | 38 | 48 | 0.38 | 0.57 | 16.459 | 10.186 |
| 6 | 4.6100 | 3.0800 | 5.9008 | 3.9424 | 0.250 | 0.890 | 55 | 67 | 46 | 56 | 50 | 61 | 0.56 | 0.86 | 10.239 | 6.809 |
| 10 | 3.0800 | 1.8300 | 3.9424 | 2.3424 | 0.310 | 0.084 | 68 | 89 | 57 | 75 | 64 | 83 | 0.94 | 1.43 | 6.842 | 4.038 |
| 16 | 1.9100 | 1.1500 | 2.4448 | 1.472 | 0.360 | 0.081 | 89 | 115 | 74 | 96 | 83 | 108 | 1.50 | 2.29 | 4.194 | 2.550 |
| 25 | 1.2000 | 0.7270 | 1.5360 | 0.93056 | 0.410 | 0.081 | 114 | 147 | 95 | 122 | 109 | 140 | 2.35 | 3.58 | 2.671 | 1.612 |
| 35 | 0.8680 | 0.5240 | 1.1110 | 0.67072 | 0.470 | 0.079 | 136 | 176 | 113 | 146 | 133 | 172 | 3.29 | 5.01 | 1.927 | 1.165 |
| 50 | 0.6410 | 0.3870 | 0.8205 | 0.49536 | 0.500 | 0.078 | 161 | 208 | 134 | 173 | 162 | 208 | 4.70 | 7.15 | 1.430 | 0.866 |
| 70 | 0.4400 | 0.2680 | 0.5632 | 0.34304 | 0.530 | 0.074 | 197 | 253 | 164 | 211 | 204 | 262 | 6.58 | 10.01 | 0.992 | 0.606 |
| 95 | 0.3200 | 0.1930 | 0.4096 | 0.24704 | 0.610 | 0.073 | 235 | 302 | 196 | 252 | 251 | 322 | 8.93 | 13.59 | 0.723 | 0.446 |
| 120 | 0.2530 | 0.1530 | 0.3238 | 0.19584 | 0.630 | 0.072 | 266 | 340 | 222 | 284 | 287 | 368 | 11.28 | 17.16 | 0.577 | 0.363 |
| 150 | 0.2060 | 0.1240 | 0.2637 | 0.15872 | 0.640 | 0.073 | 296 | 379 | 248 | 317 | 328 | 419 | 14.10 | 21.45 | 0.476 | 0.304 |
| 185 | 0.1640 | 0.0991 | 0.2099 | 0.126848 | 0.650 | 0.072 | 335 | 425 | 281 | 357 | 379 | 482 | 17.39 | 26.46 | 0.386 | 0.254 |
| 240 | 0.1250 | 0.0754 | 0.1600 | 0.096512 | 0.660 | 0.071 | 385 | 486 | 324 | 409 | 448 | 566 | 22.56 | 34.32 | 0.307 | 0.211 |
| 300 | 0.1000 | 0.0601 | 0.1280 | 0.076928 | 0.670 | 0.071 | 432 | 541 | 364 | 456 | 513 | 644 | 28.20 | 42.90 | 0.256 | 0.185 |
| 400 | 0.0778 | 0.0470 | 0.0996 | 0.06016 | 0.670 | 0.070 | 487 | 602 | 412 | 508 | 593 | 734 | 37.60 | 57.20 | 0.214 | 0.165 |
| 500 | 0.0605 | 0.0366 | 0.0774 | 0.046848 | 0.690 | 0.070 | 548 | 665 | 463 | 562 | 683 | 831 | 47.00 | 71.50 | 0.185 | 0.151 |
| 630 | 0.0469 | 0.0283 | 0.0600 | 0.036224 | 0.730 | 0.070 | 612 | 728 | 518 | 616 | 784 | 936 | 59.22 | 90.09 | 0.164 | 0.141 |

# XLPE CABLE - THREE CORE



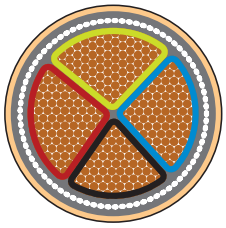
Three Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

## Physical Data

| Nominal Area | Minium No. of Strand in Conductor | | Insulation Thickness (Nominal) | Minimum inner sheath thickness | Armour Wire Dimension | | Outer Sheath Thickness | | Approx. Overall Diameter | | Approx. Net Weight of the Cables | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Round | Formed | Arm | Un-Arm | Arm | Un-Arm | Armoured | | UnArmoured | |
| Sq. mm | Aluminium Nos | Copper Nos | mm | mm | mm | mm | mm | mm | mm | mm | Kg/Km Al | Kg/Km Cu | Kg/Km Al | Kg/Km Cu |
| 4 | 1/3 | 1/7 | 0.70 | 0.30 | 1.40 | -- | 1.24 | 1.80 | 15.00 | 13.50 | 445 | 520 | 170 | 245 |
| 6 | 1/3 | 1/7 | 0.70 | 0.30 | 1.40 | -- | 1.24 | 1.80 | 17.00 | 14.50 | 505 | 615 | 210 | 315 |
| 10 | 1/7 | 1/7/6 | 0.70 | 0.30 | 1.40 | -- | 1.24 | 1.80 | 19.00 | 16.50 | 605 | 790 | 260 | 440 |
| 16 | 6 | 6 | 0.70 | 0.30 | -- | 4.0X0.8 | 1.24 | 1.80 | 18.00 | 17.00 | 560 | 850 | 335 | 625 |
| 25 | 6 | 6 | 0.90 | 0.30 | -- | 4.0X0.8 | 1.40 | 2.00 | 21.00 | 20.00 | 720 | 1170 | 480 | 925 |
| 35 | 6 | 6 | 0.90 | 0.30 | -- | 4.0x0.8 | 1.40 | 2.00 | 23.00 | 22.00 | 890 | 1525 | 600 | 1235 |
| 50 | 6 | 6 | 1.00 | 0.30 | -- | 4.0x0.8 | 1.40 | 2.00 | 25.00 | 24.50 | 1055 | 1910 | 750 | 1605 |
| 70 | 12 | 12 | 1.10 | 0.40 | -- | 4.0x0.8 | 1.56 | 2.20 | 30.00 | 28.50 | 1400 | 2635 | 1030 | 2265 |
| 95 | 15 | 15 | 1.10 | 0.40 | -- | 4.0x0.8 | 1.56 | 2.20 | 32.00 | 31.00 | 1705 | 3415 | 1290 | 3000 |
| 120 | 15 | 18 | 1.20 | 0.40 | -- | 4.0x0.8 | 1.56 | 2.20 | 35.00 | 34.00 | 2015 | 4185 | 1560 | 3725 |
| 150 | 15 | 18 | 1.40 | 0.50 | -- | 4.0x0.8 | 1.72 | 2.20 | 39.00 | 38.00 | 2415 | 5095 | 1920 | 4600 |
| 185 | 30 | 30 | 1.60 | 0.50 | -- | 4.0x0.8 | 1.88 | 2.20 | 43.00 | 42.00 | 2930 | 6265 | 2375 | 5710 |
| 240 | 30 | 34 | 1.70 | 0.60 | -- | 4.0x0.8 | 2.04 | 2.80 | 49.00 | 47.50 | 3635 | 7975 | 3025 | 7365 |
| 300 | 30 | 34 | 1.80 | 0.60 | -- | 4.0x0.8 | 2.04 | 2.80 | 49.00 | 47.50 | 3635 | 7975 | 3025 | 7365 |
| 400 | 53 | 53 | 2.00 | 0.70 | -- | 4.0x0.8 | 2.50 | 3.20 | 60.00 | 58.50 | 5450 | 1250 | 4660 | 11740 |
| 500 | 53 | 53 | 2.20 | 0.70 | -- | 4.0x0.8 | 2.68 | 3.60 | 67.00 | 65.50 | 6720 | 15760 | 5885 | 14930 |
| 630 | 53 | 53 | 2.40 | 0.70 | -- | 4.0x0.8 | 2.84 | 3.80 | 74.00 | 72.50 | 8220 | 20005 | 7320 | 19105 |

## Electrical Data

| Nominal Conductor Data | Max. DC Resisance of Conductor at 20°C | | Approx. AC Resistance of Max Operating Temp 90°C | | Appox. Capacitance | Aprox. Reactance at 50 Hz | Current Rating | | | | | | Short Circuit Rating for 1 Sec | | Volrage Drop | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | In Ground | | In Duct | | In Air | | | | | |
| Sq. mm | Ω/Km Al | Ω/Km Cu | Ω/Km Al | Ω/Km Cu | µF/Km | Ω/Km | Amps Al | Amps Cu | Amps Al | Amps Cu | Amps Al | Amps Cu | kA (rms) Al | kA (rms) Cu | V/A/Km Al | V/A/Km Cu |
| 4 | 7.4100 | 4.6100 | 9.4848 | 5.9008 | 0.065 | 0.093 | 35 | 45 | 30 | 38 | 32 | 41 | 0.38 | 0.57 | 16.459 | 10.186 |
| 6 | 4.6100 | 3.0800 | 5.9008 | 3.9424 | 0.071 | 0.089 | 46 | 56 | 38 | 47 | 42 | 52 | 0.56 | 0.86 | 10.239 | 6.809 |
| 10 | 3.0800 | 1.8300 | 3.9424 | 2.3424 | 0.081 | 0.084 | 57 | 74 | 48 | 62 | 54 | 70 | 0.94 | 1.43 | 6.842 | 4.038 |
| 16 | 1.9100 | 1.1500 | 2.4448 | 1.472 | 0.088 | 0.081 | 74 | 95 | 61 | 79 | 69 | 89 | 1.50 | 2.29 | 4.194 | 2.550 |
| 25 | 1.2000 | 0.7270 | 1.5360 | 0.93056 | 0.089 | 0.081 | 95 | 122 | 79 | 102 | 93 | 119 | 2.35 | 3.58 | 2.671 | 1.612 |
| 35 | 0.8680 | 0.5240 | 1.1110 | 0.67072 | 0.096 | 0.079 | 114 | 146 | 94 | 122 | 114 | 147 | 3.29 | 5.01 | 1.927 | 1.165 |
| 50 | 0.6410 | 0.3870 | 0.8205 | 0.49536 | 0.098 | 0.078 | 134 | 173 | 112 | 144 | 138 | 179 | 4.70 | 7.15 | 1.430 | 0.866 |
| 70 | 0.4400 | 0.2680 | 0.5632 | 0.34304 | 0.100 | 0.074 | 164 | 212 | 137 | 177 | 175 | 226 | 6.58 | 10.01 | 0.992 | 0.606 |
| 95 | 0.3200 | 0.1930 | 0.4096 | 0.24704 | 0.110 | 0.073 | 197 | 254 | 164 | 212 | 216 | 279 | 8.93 | 13.59 | 0.723 | 0.446 |
| 120 | 0.2530 | 0.1530 | 0.3238 | 0.19584 | 0.110 | 0.072 | 223 | 287 | 187 | 240 | 249 | 320 | 11.28 | 17.16 | 0.577 | 0.363 |
| 150 | 0.2060 | 0.1240 | 0.2637 | 0.15872 | 0.110 | 0.073 | 249 | 321 | 209 | 269 | 284 | 365 | 14.10 | 21.45 | 0.476 | 0.304 |
| 185 | 0.1640 | 0.0991 | 0.2099 | 0.126848 | 0.110 | 0.072 | 282 | 362 | 238 | 304 | 329 | 422 | 17.39 | 26.46 | 0.386 | 0.254 |
| 240 | 0.1250 | 0.0754 | 0.1600 | 0.096512 | 0.110 | 0.071 | 327 | 418 | 276 | 352 | 392 | 500 | 22.56 | 34.32 | 0.307 | 0.211 |
| 300 | 0.1000 | 0.0601 | 0.1280 | 0.076928 | 0.120 | 0.071 | 369 | 469 | 312 | 696 | 452 | 574 | 28.20 | 42.90 | 0.256 | 0.185 |
| 400 | 0.0778 | 0.0470 | 0.0996 | 0.06016 | 0.120 | 0.070 | 420 | 528 | 356 | 447 | 526 | 664 | 37.60 | 57.20 | 0.214 | 0.165 |
| 500 | 0.0605 | 0.0366 | 0.0774 | 0.046848 | 0.120 | 0.070 | 478 | 593 | 412 | 511 | 612 | 760 | 47.00 | 71.50 | 0.185 | 0.151 |
| 630 | 0.0469 | 0.0283 | 0.0600 | 0.036224 | 0.120 | 0.070 | 542 | 661 | 468 | 571 | 712 | 870 | 59.22 | 90.09 | 0.164 | 0.141 |



# XLPE CABLE - THREE & HALF CORE

Three & Half Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

## Physical Data

| Nominal Area | | Minium No. of Strand in Conductor | | Insulation Thickness (Nominal) | | Inner Sheath Thickness (Min) | Armour Dimension | Outer Sheath Thickness | | Approx. Overall Diameter | | Approx. Net Weight of the Cables | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Power | Neutral | | Formed | Arm | Un-Arm | Arm | Un-Arm | Armoured | | UnArmoured | |
| Sq. mm | Sq. mm | Aluminium | Copper | Sq. mm | Sq. mm | mm | mm | mm | mm | mm | mm | Kg/Km | Kg/Km | Kg/Km | Kg/Km |
| | | Nos | Nos | | | | | | | | | Al | Cu | Al | Cu |
| 25 | 16 | 6 | 6 | 0.90 | 0.70 | 0.30 | 4.0x0.8 | 1.40 | 2.00 | 22.50 | 21.50 | 805 | 1350 | 540 | 1080 |
| 35 | 16 | 6 | 6 | 0.90 | 0.70 | 0.30 | 4.0x0.8 | 1.40 | 2.00 | 24.50 | 35.50 | 980 | 1705 | 650 | 1380 |
| 50 | 25 | 6 | 6 | 1.00 | 0.90 | 0.30 | 4.0x0.8 | 1.40 | 2.00 | 27.00 | 26.00 | 1200 | 2205 | 830 | 1835 |
| 70 | 35 | 12 | 12 | 1.10 | 0.90 | 0.400 | 4.0x0.8 | 1.56 | 2.20 | 31.00 | 30.00 | 1550 | 2995 | 1145 | 1590 |
| 95 | 50 | 15 | 15 | 1.10 | 1.00 | 0.400 | 4.0X0.8 | 1.56 | 2.20 | 31.50 | 33.50 | 925 | 3920 | 1455 | 3450 |
| 120 | 70 | 15 | 18 | 1.20 | 1.10 | 0.400 | 4.0x0.8 | 1.72 | 2.20 | 38.50 | 37.00 | 2335 | 4910 | 1795 | 4370 |
| 150 | 70 | 15 | 18 | 1.40 | 1.10 | 0.500 | 4.0x0.8 | 1.72 | 2.40 | 42.00 | 41.00 | 2730 | 5820 | 2165 | 5855 |
| 185 | 95 | 30 | 30 | 1.60 | 1.10 | 0.500 | 4.0x0.8 | 1.88 | 2.60 | 46.50 | 45.50 | 3325 | 7230 | 2700 | 6610 |
| 240 | 120 | 30 | 34 | 1.70 | 1.20 | 0.600 | 4.0x0.8 | 2.04 | 2.80 | 52.50 | 51.50 | 4115 | 9180 | 3445 | 8505 |
| 300 | 150 | 30 | 34 | 1.80 | 1.40 | 0.600 | 4.0x0.8 | 2.20 | 3.00 | 87.50 | 86.50 | 4960 | 11350 | 4205 | 10595 |
| 400 | 185 | 53 | 53 | 2.00 | 1.60 | 0.700 | 4.0x0.8 | 2.52 | 3.40 | 65.50 | 65.00 | 6230 | 14420 | 5365 | 13560 |
| 500 | 240 | 53 | 53 | 2.20 | 1.70 | 0.700 | 4.0x0.8 | 2.68 | 3.60 | 72.50 | 72.00 | 7655 | 18145 | 6725 | 17215 |
| 630 | 300 | 53 | 53 | 2.40 | 1.80 | 0.700 | 4.0x0.8 | 3.00 | 4.00 | 82.00 | 81.00 | 9485 | 23105 | 8465 | 22085 |

## Electrical Data

| Nominal Conductor Data | Max. DC Resisance of Conductor at 20°C | | Approx. AC Resistance of Max Operating Temp 90°C | | Appox. Capacitance | Aprox. Reactance at 50 Hz | Current Rating | | | | | | Short Circuit Rating for 1 Sec | | Volrage Drop | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | In Ground | | In Duct | | In Air | | | | | |
| Sq. mm | Ω/Km | Ω/Km | Ω/Km | Ω/Km | | | Amps | Amps | Amps | Amps | Amps | Amps | kA (rms) | | V/A/Km | |
| | Al | Cu | Al | Cu | µF/Km | Ω/Km | Al | Cu | Al | Cu | Al | Cu | Al | Cu | Al | Cu |
| 25 | 1.2000 | 0.7270 | 1.5360 | 0.93056 | 0.089 | 0.081 | 95 | 122 | 79 | 102 | 93 | 119 | 2.35 | 3.58 | 2.671 | |
| 35 | 0.8680 | 0.5240 | 1.1110 | 0.67072 | 0.096 | 0.079 | 114 | 146 | 94 | 122 | 114 | 147 | 3.29 | 5.01 | 1.927 | 1.165 |
| 50 | 0.6410 | 0.3870 | 0.8205 | 0.49536 | 0.098 | 0.078 | 134 | 173 | 112 | 144 | 138 | 179 | 4.70 | 7.15 | 1.430 | 0.866 |
| 70 | 0.4400 | 0.2680 | 0.5632 | 0.34304 | 0.100 | 0.074 | 164 | 212 | 137 | 177 | 175 | 226 | 6.58 | 10.01 | 0.992 | 0.606 |
| 95 | 0.3200 | 0.1930 | 0.4096 | 0.24704 | 0.110 | 0.073 | 197 | 254 | 164 | 212 | 216 | 279 | 8.93 | 13.59 | 0.723 | 0.446 |
| 120 | 0.2530 | 0.1530 | 0.3238 | 0.19584 | 0.110 | 0.072 | 223 | 287 | 187 | 240 | 249 | 320 | 11.28 | 17.16 | 0.577 | 0.363 |
| 150 | 0.2060 | 0.1240 | 0.2637 | 0.15872 | 0.110 | 0.073 | 249 | 321 | 209 | 269 | 284 | 365 | 14.10 | 21.45 | 0.476 | 0.304 |
| 185 | 0.1640 | 0.0991 | 0.2099 | 0.126848 | 0.110 | 0.072 | 282 | 362 | 238 | 304 | 329 | 422 | 17.39 | 26.46 | 0.386 | 0.254 |
| 240 | 0.1250 | 0.0754 | 0.1600 | 0.096512 | 0.110 | 0.071 | 327 | 418 | 276 | 352 | 392 | 500 | 22.56 | 34.32 | 0.307 | 0.211 |
| 300 | 0.1000 | 0.0601 | 0.1280 | 0.076928 | 0.120 | 0.071 | 369 | 469 | 312 | 696 | 452 | 574 | 28.20 | 42.90 | 0.256 | 0.185 |
| 400 | 0.0778 | 0.0470 | 0.0996 | 0.06016 | 0.120 | 0.070 | 420 | 528 | 356 | 447 | 526 | 664 | 37.60 | 57.20 | 0.214 | 0.165 |
| 500 | 0.0605 | 0.0366 | 0.0774 | 0.046848 | 0.120 | 0.070 | 478 | 593 | 412 | 511 | 612 | 760 | 47.00 | 71.50 | 0.185 | 0.151 |
| 630 | 0.0469 | 0.0283 | 0.0600 | 0.036224 | 0.120 | 0.070 | 542 | 661 | 468 | 571 | 712 | 870 | 59.22 | 90.09 | 0.164 | 0.141 |